Team Meetings

Week 9

Tuesday, Mar. 3rd, 2026 (Studio 9A):

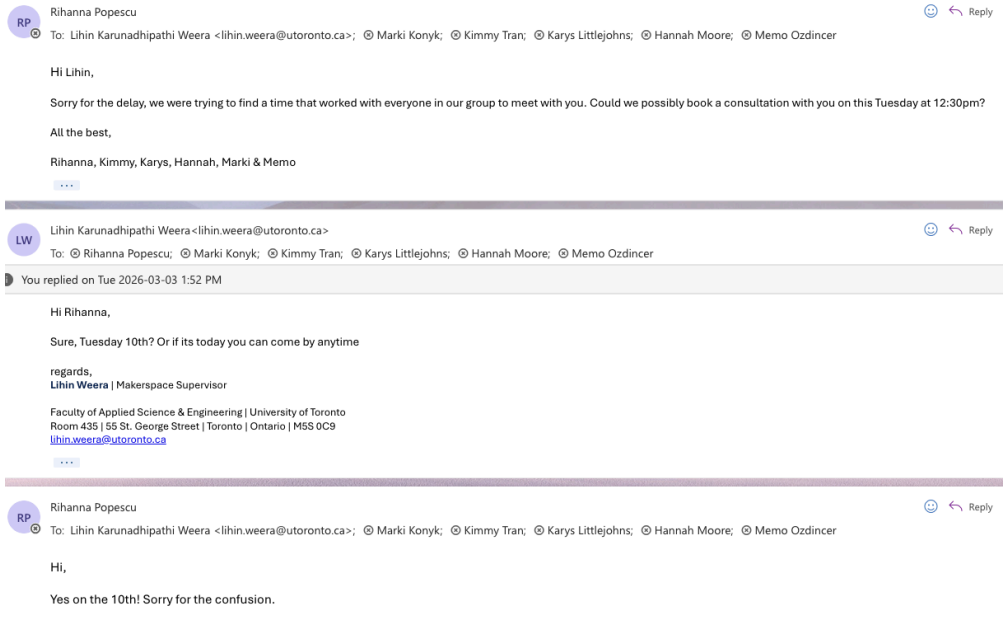
Meeting Agenda

- Email MYFab Mentor Consultation to book for Tuesday
- Make Prototyping Testing house-based

Meeting Minutes

members present: Rihanna, Karys, Hannah, Memo, Marki, Kimmy

- Emailed MYFab Mentor and booked a consultation for the following Tuesday



Rihanna Popescu
To: Lihin Karunadhipathi Weera <lihin.weera@utoronto.ca>; Marki Konyk; Kimmy Tran; Karys Littlejohns; Hannah Moore; Memo Ozdincer

Hi Lihin,

Sorry for the delay, we were trying to find a time that worked with everyone in our group to meet with you. Could we possibly book a consultation with you on this Tuesday at 12:30pm?

All the best,

Rihanna, Kimmy, Karys, Hannah, Marki & Memo

Lihin Karunadhipathi Weera<lihin.weera@utoronto.ca>
To: Rihanna Popescu; Marki Konyk; Kimmy Tran; Karys Littlejohns; Hannah Moore; Memo Ozdincer

You replied on Tue 2026-03-03 1:52 PM

Hi Rihanna,

Sure, Tuesday 10th? Or if its today you can come by anytime

regards,
Lihin Weera | Makerspace Supervisor

Faculty of Applied Science & Engineering | University of Toronto
Room 435 | 55 St. George Street | Toronto | Ontario | M5S 0C9
lihin.weera@utoronto.ca

Rihanna Popescu
To: Lihin Karunadhipathi Weera <lihin.weera@utoronto.ca>; Marki Konyk; Kimmy Tran; Karys Littlejohns; Hannah Moore; Memo Ozdincer

Hi,

Yes on the 10th! Sorry for the confusion.

- *Rihanna* brought up that before committing to any design direction, the team needed to know what materials were available at MYFab
- **Key Decision: Ridge track to be prioritized first.** The team agreed that the ridge track needed to come first since all other subsystems depended on it being functional

Action Items

- Research potential prototyping materials for the ridge track system or 3D printing designs → **all members individually before Studio 10A**

Thursday, Mar. 5th, 2026 (Studio 9B):

Meeting Agenda

- Discuss prototype system next steps
- Organize team schedule

Meeting Minutes

members present: Rihanna, Karys, Hannah, Memo, Marki, Kimmy

- Created prototype design document to add notes for the two prototypes

Action Items

- Review prototype design document and add individual notes
- Begin materials list for ridge track system

Week 10

Monday, Mar. 9th, 2026 (Lecture 10A):

Meeting Agenda

- Create ridge track materials lists with details and specifications

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy

- Started researching materials on MYFab to be used for ridge track prototype design (added to prototype design folder)

Action Items

- Meet with MYFab Mentor to discuss ridge track system → ***as a team*** during ***Studio 10A***
- Add to procurement document → ***as a team*** by ***Lecture 10B*** on Wednesday

Tuesday, Mar. 10th, 2026 (Studio 10A):

Meeting Agenda

- Meet with MYFab Mentor to discuss ridge track system
- Update Ridge Track Materials Lists with details and specifications

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Spoke with MYFab Mentor about potential prototyping materials and methods
- Settled on aluminum ridge track system
- Started procurement document

Action Items

- Add to procurement document → ***as a team*** by ***Lecture 10B*** on Wednesday
- Order parts from MYFab → ***as a team*** by ***Lecture 10B*** on Wednesday

Wednesday, Mar. 11th, 2026 (Lecture 10B):

Meeting Agenda

- Add materials to procurement document
- Submit order for parts on MYFab

Meeting Minutes

members present: Hannah, Marki, Kimmy

- Submitted Order #1 for parts on MYFab; aluminum extrusions, dc motor, limit switch, 4 hole joining strip, nuts, screws
- Added to procurement document
- Discussed Nozzle + Nozzle to Cart Attachment

Action Items

- CAD Nozzle + Nozzle to Cart Attachment
- Assemble mechanical parts
- Build Electrical and Software system for controlling motor

Thursday, Mar. 12th, 2026 (Studio 10B):

Meeting Agenda

- Discuss and Design CAD model for Nozzle + Nozzle to Cart Attachment

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Discussed CAD model for Nozzle + Nozzle to Cart Attachment
- Submitted Order #2 for additional parts on MYFab; bearings, screws, nuts
- Submitted 1st 3D print order for the pulley system
- Added to procurement document

Action Items

- Create testing protocols → ***as a team*** on ***Lecture 11A*** (Monday, Wednesday, or Friday?)
- Build initial prototype (Ridge Track System) → ***as a team*** on ***Studio 11A*** (Tuesday)
- Build Electrical and Software system for controlling motor → ***as a team*** on ***Lecture 11B*** (Wednesday)
- Test prototype Monday, March 23rd

Week 11

Tuesday, Mar. 17th, 2026 (Studio 11A):

Meeting Agenda

- Assemble Ridge Track Prototype
- Finalize CAD for timing belt
- Start thinking about the electrical and software systems
- Look at CAD models for Air Nozzle

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Assembled Ridge Track
- Went to MYFab; connected pieces together used hot glue to connect bearings to the 3D printed pulley system
- Created Electrical Circuit and Process & Decisions document
- Created document for List of Changes During Integration
- Looked at Air Nozzle CAD's available online for reference
- Subsystem Check-In with TA's

Action Items

- Build initial prototype (Ridge Track System) → ***as a team*** on ***Studio 11A*** (Tuesday)

- Build Electrical and Software system for controlling motor → ***as a team*** on ***Studio 11B***
- Create testing protocols
- Test prototype

Thursday, Mar. 19th, 2026 (Studio 11B):

Meeting Agenda

- Attach timing belt to system
- Start thinking about the electrical and software systems
- CAD model for Air Nozzle
- Create Specification for Prototype System sheet

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Created Specification for Prototype System sheet; included objectives for each subsystem
- Dimensions and research for Air Nozzle CAD
- Decided on prototyping longer system to highlight traversal of roof
- Added to List of Changes During Integration during the integration of the ridge track and nozzle carriage
-
- Ordered additional materials; aluminum extrusion (decided on longer prototype), screws, nuts, joining strips

Action Items

- Create testing protocols → ***as a team*** on ***Lecture 12C*** (Monday, Wednesday, or Friday?)
- Build Electrical and Software system for controlling motor → in ***Lecture 12A***

Week 12

Tuesday, Mar. 24th, 2026 (Studio 12A):

Meeting Agenda

- Build electrical and software system for controlling motor
- Continue CAD work for Air Nozzle and cart attachment

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Worked on electrical and software system for motor control
- Continued CAD work for Air Nozzle and nozzle-to-cart attachment
- Reviewed components needed for stepper motor integration
- **Switched from a timed DC motor to a Nema 17 Stepper Motor with an A4988 driver**
- **Time-based dead-reckoning is inaccurate due to friction and voltage drops. Steppers provide absolute millimeter-level precision**
- **All members were involved in making this decision**

Action Items

- Order stepper motor and driver components from MYFab → ***as a team*** by Studio 12B (Thursday)
- Continue the electrical and software system → ***as a team*** on Studio 12B

Thursday, Mar. 26th, 2026 (Studio 12B):

Meeting Agenda

- Order stepper motor components from MYFab
- Continue electrical and software system
- Discuss testing protocols

Meeting Minutes

members present: Rihanna, Hannah, Marki, Kimmy, Karys, Memo

- Ordered stepper motor nema, driver A4988, dc barrel adapter, motor driver from MYFab
- Added materials to procurement document
- Updated Specification for Prototype System sheet; justifications for each objective
- Added to List of Changes During Integration during the integration of the ridge track and nozzle carriage
- Need to add more friction to axle pulley
- Connect nozzle cart to timing belt
- Increase friction between the motor axle and pulley axle for better movement of belt

Action Items

- Set up stepper motor (NEMA) with A4988 driver → ***as a team*** by Saturday, March 28th
- Continue integration of electrical + software and structural → ***as a team*** by Saturday, March 28th
- Create testing protocols → ***as a team*** by Saturday, March 28th
- Run tests → ***as a team*** by Sunday, March 29th

– MARCH 22nd LARGE ORDER DEADLINE
– MARCH 30th DESIGN DOSSIER #2 DEADLINE

- ☐ **Prototype Overview**
- ☐ **Specification for Prototype System**
- ☐ **Verification Protocol & Results Spreadsheet**
- ☐ **Process Artifacts for DESIGN**
- ☐ **Process Artifacts for subsystem fabrication**
- ☐ **Process Artifacts for Integration**
- ☐ **Process Artifacts for Verification**
- ☐ **Visual Abstract**
- ☐ **UPDATE TEAM MEETINGS + PROJECT WORK LOGS**

4.1 Visual Abstract

Due Date: Monday, March 30 at 11:59 **AM**

Your group should submit a Visual Abstract (VA) which summarizes your Prototype System. Its purpose is to provide a high-level summary of the outcomes of your prototyping work, which may be shared with Opportunity Champions to bring the ideas you generate beyond the course.

Your Visual Abstract must include the following elements:

- ☐ A photograph or photographs of the completed Prototype System.
- ☐ A depiction of the key features of the Prototype System.
- ☐ An assessment of the function(s) and/or performance of the Prototype System relative to key elements of the specification your team developed.
- ☐ At least one representation of how the Prototype System relates to the overall Design Concept (i.e., a clear representation of which aspect(s) of the Design Concept were realized in the Prototype System, and how this realization differs from the full Design Concept).
- ☐ At least one representation of how your Design Concept fits within the context of your chosen Opportunity Space.

Project Work Logs

Work Completed: Ordered materials; and added to procurement document
Date/Time: March 11th
Members: Kimmy, Hannah, Marki
Artifacts: [MYFab Order Screenshots](), *Procurement and Budget Tracker*

Clear Form

MYFab Purchase order form

Screenshot Purchase order (jpg)

UtorID trankimm

Total (CAD) $12.66

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
Switch SPST (limit switch)		2	0.226	(blank)	288 MECHATRONICS	Rental	0.452
2550 T-Slotted Aluminum Extrusion		2	1.442	80	141 MATERIALS	Rental	2.884
Mini Ball Bearings		2	1.2	100	6 HARDWARE	Purchase	2.4
DC Gear motor 3-7.2V 210 RPM		1	0.47	(blank)	287 MECHATRONICS	Rental	0.47
4 Hole Joining Strip		2	2.316	30	236 MATERIALS	Rental	4.632
25 S M5 x 0.8 T-NUT		8	0.11	1000	163 MATERIALS	Rental	0.88
Hex Button Head Screw M5 x pitch	0.8 - 10 mm	8	0.05	500	196 MATERIALS	Rental	0.4

C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
Material Cost Table														
Material	Rental	Rental Cost	Purchase Cost	Number Needed	Link to Find	Total Cost Rental	Total Cost Purchase	Purchase or Rental	Total Cost		<-- assumes rental over purchase			
Aluminum T-Slot	Yes	1.442	7.21	3		4.326	21.63	Rental	4.326			Purchase		
210 rpm DC Motor	Yes	0.47	2.35	1		0.47	2.35	Rental	0.47			Rental		
SPST Switch (limit switch)	Yes			2		0	0	Rental	0					
						0	0		0					
Ball Bearing (for wheels)				2		0	0		0					
Wheels						0	0		0					
						0	0		0					
Belt	later					0	0		0					
Air Pumps	later					0	0		0					
						0	0		0					
Total									4.796					

Work Completed: Submitted Order #2 for additional parts on MYFab; bearings, screws, nuts, 3D print order, and added to procurement document
Date/Time: March 12th
Members: Kimmy, Hannah
Artifacts:[MYFab order screenshots](), *Procurement and Budget Tracker*

UtorID trankimm

Total (CAD) $0.64

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
25 S M5 x 0.8 T-NUT		4	0.11	1000	163 MATERIALS	Rental	0.44
Hex Button Head Screw M5 x pitch	0.8 - 10 mm	4	0.05	500	196 MATERIALS	Rental	0.2

UtorID trankimm

Total (CAD) $2.32

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
Bearing (5mm ID x 12mm OD, 4mm	Thickness)	2	0.568	29	336 MECHATRONICS	Rental	1.136
25 S M5 x 0.8 T-NUT		4	0.11	1000	163 MATERIALS	Rental	0.44
Hex Button Head Screw M5 x pitch	0.8 - 10 mm	4	0.05	500	196 MATERIALS	Rental	0.2

Material	Parent Project	Rental option?	Rental Cost	Purchase Cost	Number Needed	Link to Find	# Total Cost Rental	# Total Cost Purchase	Purchase or Rental	# Total Cost
Aluminum T-Slot	Ridge Track	Yes	$1.44	$7.21	2		2.884	14.42	Rental	2.884
210 rpm DC Motor	Ridge Track	Yes	$0.47	$2.35	1		0.47	2.35	Rental	0.47
SPST Switch (limit switch)	Ridge Track	Yes	$0.23		2		0.452	0	Rental	0.452
Bearings	Ridge Track	Yes	$0.57		4		2.272	0	Rental	2.272
Nut	Ridge Track	Yes	$0.11		16		1.76	0	Rental	1.76
Screw	Ridge Track	Yes	$0.05		16		0.8	0	Rental	0.8
4-Hole Joining Strip	Ridge Track	Yes	$2.32		2		4.632	0	Rental	4.632
Belt	Ridge Track	later					0	0		0
	N/A									
	N/A									
Total										13.27

Print Name	Parent Project	Material	Time to Print	Cost to Print
Axles and Attachments V1	Ridge Track	PLA	2h14m	$1.34
Track V1	Ridge Track			
	N/A			
	N/A			
Total				$1.34

Type of Cost	$
Material Costs	$13.27
Printing Costs	$1.34
Budget Total	$175.00
Total Costs	$14.61
Remaining Budget	$160.39

Work Completed: Ridge Track Prototype - CAD of pulleys (Version 1)
Date/Time: March 14th
Members:
Artifacts:

Work Completed: Ridge Track Prototype - CAD of timing belts
Date/Time: March 15th
Members:
Artifacts:

Work Completed: Ridge Track Prototype - 3D pulleys & timing belts
Date/Time:
Members:
Artifacts:

Work Completed: Ordered additional materials; aluminum extrusion (decided on longer prototype), screws, nuts, joining strips
Date/Time: March 19th
Members: Kimmy, Hannah
Artifacts: [MYFab order screenshots]

Clear Form

MYFab Purchase order form

Screenshot Purchase order (jpg)

UtorID trankimm

Total (CAD) $7.89

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
2550 T-Slotted Aluminium Extrusion	20 cm	1	1.442	-	141 MATERIALS	Rental	1.442
Hex Button Head Screw M5 x pitch (	1 unit	8	0.05	(blank)	196 MATERIALS	Rental	0.4
25 S M5 x 0.8 T-NUT	1 unit	8	0.11	-	163 MATERIALS	Rental	0.88
4 HOLE JOINING STRIP	1 unit	2	2.316	-	236 MATERIALS	Rental	4.632

Work Completed: Specification Protocol - Updated related specifications for prototype system based on subsystem objectives
Date/Time: March 19th
Members: Kimmy
Artifacts: [ESC204S_2026_DD2_Specification]

Top-Level or Subsystem Objective ID	Statement of Top-Level or Subsystem Objective	Specification ID	Statement of Related Specification for Prototype System	Justification
Electrical Subsystem				
O-1.1	Removal and/or Prevention of organic debris from simple slanted home roofs, with a larger reduction in Hazard Rating due to debris removal (see metric) preferred.	O-1.1-ELEC-1	Shall power a motor-driven system capable of moving a nozzle along the roof ridge to clear debris.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-ELEC-1	Shall allow the system to be activated and controlled without requiring access to the roof.	
O-2.2	Improve the overall risk profile to a user due to organic debris removal activities, at worst adding no new hazards.	O-2.2-ELEC-1	All electrical connections shall be insulated and protected from exposure to water during operation	
O-3.2	Minimize use of materials consumed (e.g., fuel, water, disposable parts, etc.) during operation, with no consumables preferred.	O-3.2-ELEC-1	Shall include limit switches at both ends of the track to stop/change direction of motion when endpoints are reached.	
Structural Subsystem				
O-1.4	Operate in varying weather conditions (dry, moderately wet).	O-1.4-STRU-1	Shall resist waer exposure without deformation or loss of function.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-STRU-1	Shall be maintainable without requiring a user to stand on the roof	
O-4.1	Integration does not decrease building aestethics.	O-4.1-STRU-1	Shall be low-profile and follow the contour of the roof when installed.	
O-4.2	Does not cause damage to preexisting roof structure or shingles.	O-4.2-STRU-1	Shall not damage shingles during installation or operation.	
O-3.1	Operate quietly, generating not more than 70 dBA at the nearest listener during operation [4].	O-3.1-STRU-1	There shall support components in a way that limits vibration and noise during operation.	
Software Subsystem				
O-1.3	Reduce the active time required for a user to remove organic debris near to a structure, with no active time preferred.	O-1.3-SOFT-1	Shall have a "CLEARING" state where the system traverses the full track across the roof ridge.	
O-1.4	Operate in varying weather conditions (dry, moderately wet).	O-1.4-SOFT-1	Shall not enter "CLEARING" state if unsafe conditions (e.g., excessive mositure) are detected.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-SOFT-1	There shall be a "CLEARING" state where the nozzle on the track traverses the ridge of the roof.	
O-3.2	Minimize use of materials consumed (e.g., fuel, water, disposable parts, etc.) during operation, with no consumables preferred.	O-3.2-SOFT-1	Shall be an "OFF" state where all motors and actuators are inactive.	

Work Completed: Ridge Track Prototype - CAD of pulleys (Version 2)
Date/Time: March 22nd
Members:
Artifacts:

Work Completed: Ridge Track Prototype - 3D printing of pulleys (Version 2) & attachments
Date/Time: March 22nd
Members:
Artifacts:

Work Completed: Electrical Subsystem Version 1 (moving the motor)
Date/Time: March 24th
Members:
Artifacts:

Work Completed: Team Charter Updates
Date/Time: March
Members:
Artifacts:

Work Completed: Ordered final materials; stepper motor nema, driver A4988, dc barrel adapter, motor driver from MYFab
Date/Time: March 26th
Members: Kimmy, Memo
Artifacts: [MYFab order screenshots]

UtorID: frankimm — Total (CAD): $2.46

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
DRV8833 Motor Driver		1	1.924	120	127 MECHATRONICS	Rental	1.924

UtorID: mehmet — Total (CAD): $7.56

Quantity is limited to 20 of each item

Description	Specs	Quantity	Unit $	Out of Stock	ID	Terms	Subtotal
Stepper motor Nema 17		1	4.346	(blank)	48 MECHATRONICS	Rental	4.346
Stepper Motor Driver A4988	Motor drivers	1	2.344	(blank)	114 MECHATRONICS	Rental	2.344
DC Barrel Adapter - BrdBrd		1	0.326	(blank)	63 MECHATRONICS	Rental	0.326

Work Completed: Specification Protocol - Updated justifications for each specification based on requirement of system
Date/Time: March 27th
Members: Kimmy
Artifacts: [ESC204S_2026_DD2_Specification]

Top-Level or Subsystem Objective ID	Statement of Top-Level or Subsystem Objective	Specification ID	Statement of Related Specification for Prototype System	Justification for Specification	Notes
		Electrical Subsystem			
O-1.1	Removal and/or Prevention of organic debris from simple slanted home roofs, with a larger reduction in Hazard Rating due to debris removal (see metric) preferred.	O-1.1-ELEC-1	The motor-driven ridge track system with an attached nozzle carriage shall clear 75% of debris from the slanted roof.	To validate whether a motor driven system can reliably traverse a roof ridge to clear debris with the nozzle carriage cart on top. This is something that cannot be confirmed through analysis alone due to variability in friction and motor behaviour. A failure in this specification indicates if the issue lies in power, friction or track geometry.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-ELEC-1	The motor-driven ridge track system with an attached nozzle carriage shall be activated and controlled without requiring access to the roof.	This prototype design uses ground level control and its reliability must be verified to ensure it actually eliminates the hazards present to the user during operation. A failure would indicate the control architecture needs to be reconsidered as the safety of the user depends entirely on remote operability.	
O-2.2	Improve the overall risk profile to a user due to organic debris removal activities, at worst adding no new hazards.	O-2.2-ELEC-1	The motor-driven ridge track system with an attached nozzle carriage shall have insulated electrical compoentes which are protected from exposure to water during operation.	Electrical connections exposed to outdoor conditions present a safety hazard if not properly protected. This must be verified on the prototype system to ensure the system does not introduce new risks to the user. A failure would indicate that the enclosure or insulation method should be revised before the system can be safely implemented.	
O-3.2	Minimize use of materials consumed (e.g., fuel, water, disposable parts, etc.) during operation, with no consumables preferred.	O-3.2-ELEC-1	The motor-driven ridge track system with an attached nozzle carriage shall include limit switches at both ends of the track to minimize the materials consumed (electrical power to traverse the ridge of the roof and to blow the air through the nozzle) by stopping/changing the direction of motion when limit switches are reached.	Limit switches at both ends of the track are the method for preventing unnecessary motor operation. The limit switches mechanism must be verified on the prototype to confirm the system minimises energy consumption during use. A failure would indicate that switch placement or firmware response needs to be adjusted to ensure the carriage movement can be controlled (stopped and reversed) at the respective endpoints.	
		Structural Subsystem			
O-1.4	Operate in varying weather conditions (dry, moderately wet).	O-1.4-STRU-1	The motor-driven ridge track system with an attached nozzle carriage shall resist water exposure without deformation of the system or loss of function of the system.	Given the prototype system would be exposed to outdoor weather during operation, its resistance to water must be verified on the prototype to confirm the system can function in moderately wet conditions. A failure would indicate that material or joining choices need to be reconsidered before the system could be implemented.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-STRU-1	The motor-driven ridge track system with an attached nozzle carriage shall be maintainable without requiring a user to stand on the roof.	Eliminating the need for roof access during maintenance is a key safety requirement, and this must be verified on the prototype to confirm the structural layout allows all servicing to be performed from ground level or a ladder. A failure would indicate that component placement needs to be revised to remove the requirement for the user to stand on the roof.	
O-3.1	Operate quietly, generating not more than 70 dBA at the nearest listener during operation [4].	O-3.1-STRU-1	The motor-driven ridge track system with an attached nozzle carriage shall not exceed 70 dBA at 3 meters from the nearest listener.	Noise generated during operation must not disturb nearby occupants, and structural vibration damping must be verified on the prototype to confirm the system stays within the 70 dBA limit. A failure would indicate that the rail mounting, carriage bearings, or motor coupling needs to be revised to reduce noise transmission.	
O-4.1	Integration does not decrease building aesthetics.	O-4.1-STRU-1	The motor-driven ridge track system with an attached nozzle carriage shall be low-profile and follow the contour of the roof when installed.	The system must not negatively impact the appearance of the building it is installed on, and its visual profile must be verified on the prototype to confirm it follows the roofline without being visually dominant. A failure would indicate that the rail height or mounting geometry needs to be adjusted to achieve a lower-profile installation.	
O-4.2	Does not cause damage to preexisting roof structure or shingles.	O-4.2-STRU-1	The motor-driven ridge track system with an attached nozzle carriage shall not damage shingles during installation or operation.	Shingle damage during installation or operation would introduce new costs and risks for the user, and the system's contact with the roof surface must be verified on the prototype to confirm no damage occurs. A failure would indicate that the contact surface design or load distribution strategy needs to be revised to protect the shingles.	
		Software Subsystem			
O-1.3	Reduce the active time required for a user to remove organic debris near to a structure, with no active time preferred.	O-1.3-SOFT-1	The motor-driven ridge track system with an attached nozzle carriage shall have a "CLEARING" state where the system traverses 2 passes of the full track across the roof ridge clearing 75% of debris from the slanted roof which can be activated by the press of a button allowing no active time spent for user to clear debris.	The CLEARING state verifies that the system completes a full debris-clearing cycle from a single user command, with no active input required during traversal. A failure would indicate the motor control or state logic needs adjustment to achieve true autonomous operation.	
O-2.1	Reduce the potential for injury to a user due to slips, trips and falls while removing organic debris near to a structure, with elimination of slip, trip and fall hazards preferred [3]. No roof access should be necessary for maintenance.	O-2.1-SOFT-1	The motor-driven ridge track system with an attached nozzle carriage shall have a "CLEARING" state where the system traverses 2 passes of the full track across the roof ridge which can be powered on by pressing a button allowing ground level control.	The CLEARING state verifies that the nozzle traverses the full ridge without requiring the user to physically access or guide anything on the roof. A failure would indicate the firmware cannot reliably complete a pass independently, reintroducing the need for manual roof-level intervention.	
O-3.2	Minimize use of materials consumed (e.g., fuel, water, disposable parts, etc.) during operation, with no consumables preferred.	O-3.2-SOFT-1	Shall be an "OFF" state where all motors are inactive.	The OFF state is the primary mechanism for ensuring no energy is consumed when the system is not in use, and its correct function must be verified on the prototype to confirm all motors are fully de-energised. A failure would indicate that the state transition logic or hardware configuration needs to be revised to prevent residual motor energisation.	

Tab 3

4.2.9 The `05_TeamworkProcess` Folder

At the end of Phase 3, your `05_TeamworkProcess` folder **must** contain the following Process and Outcome Artifacts:

- ☐ A complete Team Charter, with revisions as appropriate.
- ☐ An expanded Project Schedule for Phase 3 of ESC204, with both intended and actual timelines.
- ☐ A Procurement Tracker artifact which shows all MYFab Orders, and accounts for all components/materials used during Phase 3 (completion only).
- ☐ A Responsibility Tracker artifact which shows who was responsible for which aspect(s) of your work during Phase 3 of the course (completion only).
- ☐ The `05.0_EngineeringNotebook` folder containing a collection of artifacts which capture a chronological account of your work during Phase 3.

The Team Charter was created during Phase 2, and should be updated to reflect your team's experiences and actual processes used throughout the course. Initial updates to your Team Charter will be facilitated during Studio. Similarly, your Project Schedule and Responsibility Tracker should be updated to reflect your team's activities throughout Phase 3; these updates will be facilitated in Studio and lecture throughout the phase.

You will also prepare a Procurement Tracker which captures all of your team's MYFab orders throughout Phase 3. This artifact is assessed for completion only; however, it must be updated regularly throughout the term to ensure your team is on-budget and help your Studio TAs understand what your team has ordered.

The remainder of this section describes the **Engineering Notebook** you should continue to create as a team as you work on your project. This will be similar to the Engineering Notebook you created as part of Phase 2; however, you should **not** include all of your old entries from Phase 2 in your Phase 3 Dossier. Instead, if you'd like to reference conversations or work that you completed during Phase 2, please include these as separate artifact(s) in the `05_TeamProcess` folder, or a subfolder.

The `05.0_EngineeringNotebook` folder should contain a set of artifacts which together comprise your team's Engineering Notebook. Though there are many interpretations of what an "Engineering Notebook" can be, in this course, your team's Engineering Notebook should be a digital, shared, chronological record of all of your team's activities during Phase 3. In particular, your Engineering Notebook should include:

- ☐ A record of each team meeting including the time/date, who was present, what was discussed (i.e., meeting minutes), and any action items.
- ☐ Logs of work done on your project, including significant work completed by the whole team, by individuals and by sub-teams. Work Log entries should include the time/date work was completed, who was present, and a summary of progress (including images, sketches, photos, links to created artifacts, questions/concerns, or other material as appropriate).

Your team may choose how to organize your Engineering Notebook to suit your particular work style, but the intention is to ensure your team captures your design process as it happens which creates a primary record of your engineering work. This means all entries must include a date and time stamp, and you **should not** remove/erase/delete information once it's been entered in the Engineering Notebook. Your notebook may be a single document or multiple documents.

Though your Engineering Notebook helps capture work as it happens, you may still place artifacts you create in other folders, and link to them in the Engineering Notebook.

⊕ **EXAMPLE**: Say you and a teammate work together on fabricating a structural element for a subsystem of your Prototype System in MYFab. Photos of the work you completed could be stored in the `04.2_Build` Folder (Section 4.2.6), but notes about the work session itself and a summary of what was accomplished (a Work Log) should appear in the Engineering Notebook. You could choose to include links to the photos from the Engineering Notebook to make the entry more useful to teammates who were not at the work session.